

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

QUESTION BANK

SUBJECT NAME: COMPUTER NETWORKS

SUBJECT CODE: CST52

UNIT-I

2 MARKS

1. What is Network?
2. Define Computer Network?
3. Define Internet or internetwork?
4. What is distributed system?
5. What is meant by Middleware?
6. What is meant by World Wide Web (WWW)?
7. Mention the uses of Computer Networks?
8. What is meant by Virtual Private Networks (VPN)?
9. What is meant by Client?
10. What is meant by Server? List its types?
11. What is Client-Server model?
12. What is meant by IP telephony?
13. What is meant by Cookies?
14. What is meant by Phishing?
15. Name the two types of transmission technology? (MAY 2016, 2017)
16. What is broadcast links and point-to-point links?
17. What is meant by unicasting, broadcasting and multicasting?
18. What is meant by PAN, LAN, MAN and WAN?
19. What is meant by Access Point (AP)?
20. Define Ethernet?
21. What is the purpose of network interface card? (NOV 2016)
22. Define Switch and Router?
23. Define Protocol?
24. What is Network Architecture?
25. Define Protocol Stack?

26. What is meant by error detection and error correction?
27. What is meant by statistical multiplexing?
28. List the design issues to secure networks?
29. Define Packets?
30. Difference between Connection-oriented and Connection-less Service? (MAY 2017)
31. Difference between store-and-forward switching and cut-through switching?
32. List the various service primitives?
33. Define Internet Protocol (IP)?
34. Difference between OSI Model and TCP/IP Model?
35. What are the functions of the transport layer in the OSI reference model? (NOV 2015)
36. What is meant by TCP?
37. What is meant by UDP?
38. List the critical issues in OSI Model?
39. List the examples of computer networks?
40. Define Sockets?
41. Define multipath Fading?
42. List the various types of RFID?
43. List the various applications in sensors?
44. Define bandwidth?
45. What is meant by Nyquist Bit Rate?
46. What is meant by Shannon capacity?
47. What are Half and Full duplex mode?
48. List the various types of network topology?
49. What are the two categories of physical media?
50. Differentiate guided and unguided transmission medium? (NOV 2016)
51. What is meant by twisted pair cable?
52. What are the two types of Fibers?
53. Comparison between comparison of semiconductor diodes and LEDs?
54. Mention the advantages of Optical fiber?
55. Mention the disadvantages of Optical fiber?
56. What is unguided media?
57. What is the difference between a passive star and an active repeater in a fiber optic network? (MAY 2016)
58. Mention the various types of wireless transmission?
59. What are the standards used in wireless communication? (NOV 2015)
60. Mention the types of spread spectrum?
61. What is meant by IrDA?

62. What are the various communication satellites?
63. What are the various Frequency Bands for Satellite Communication?

11 MARKS

1. Explain in detail about the uses of Computer Networks? Explain with respective of real world applications?
2. Explain in detail about Network Hardware and types of Networks?
3. Explain in detail about Protocol Hierarchy in Network Software? (6 MARKS)
4. Explain the design issues for layers in Network Software? (5 MARKS)
5. Compare the Connection-Oriented Versus Connectionless Service? (6 MARKS)
6. Write a short on Socket Primitives? (5 MARKS)
7. Explain in detail about OSI Reference Model? (NOV 2015, 2016) (MAY 2016)
8. Explain in detail about TCP/IP Reference Model?
9. Compare and contrast the ISO/OSI reference model with the TCP/IP reference model. (MAY 2017)
10. Explain the various examples of networks?
11. Explain the theoretical basis for Data Communication? (NOV 2015)
12. Explain in detail about transmission guided media? (NOV 2015) (MAY 2016, 2017)
13. Explain in detail about wireless transmission? (MAY 2017)
14. Explain the various communication satellites in detail? (NOV 2015)
15. Explain in details the different topologies of networks with examples? (NOV 2016)

UNIT-II

2 MARKS

1. What is meant by data link layer?
2. List the functions of data link layer?
3. What is meant by frames?
4. List the issues in data link layer?
5. What are the services offered by data link layer?
6. Mention the four methods used in Framing?
7. Define byte stuffing?
8. Define bit stuffing?
9. What is the use of error control?
10. What is the necessity of flow control? (MAY 2017)
11. Mention the two approaches in flow control mechanism?
12. What is Network Interface Cards (NIC)?
13. List the various error correcting codes?
14. List the various codes in error correcting codes?
15. Define code-word?

16. What is hamming codes?
17. Define Hamming distance? (MAY 2016)
18. What is convolutional codes?
19. Mention the two type of decision decoding?
20. What is Reed-Solomon code?
21. Define LDPC?
22. Mention the various error detecting codes?
23. What is meant by parity bit?
24. Define checksum?
25. What is meant by Cyclic Redundancy Check (CRC)? (NOV 2016)
26. Define Frame header?
27. What is meant by Stop-and-Wait protocol?
28. What is meant by Automatic Repeat request (ARQ)?
29. Define piggybacking?
30. What is sliding window protocol?
31. What is meant by Go-back-N ARQ protocol?
32. What is meant by Selective Repeat ARQ protocol?
33. Define Medium Access Control (MAC)?
34. Mention the two types of channel allocation problem?
35. List the five assumptions of dynamic channel allocation?
36. Define ALOHA? Mention its versions of ALOHA?
37. Define Pure ALOHA.
38. Define Slotted ALOHA.
39. Define CSMA protocols?
40. Mention the various types of CSMA?
41. What are the difference between collision free and contention based network protocols? (NOV 2015)
42. What is a bit map protocol?
43. Define token?
44. What is meant by token ring protocol?
45. What is meant by token bus?
46. What is meant by binary count down?
47. What is meant by limited-contention protocol?
48. What is meant by hidden and exposed terminal problem?
49. Define MACA?
50. What is meant by Access Point (AP)?
51. What is meant by distribution system?

52. What is ad hoc network?
53. List the various IEEE 802.11 physical layer?
54. What is meant by CSMA/CA?
55. What are the two modes in collision avoidance?
56. What is meant by Network Allocation Vector (NAV)?
57. Mention the various Inter-Frame spacing?
58. What is meant by LLC?
59. Define OFDMA?
60. Mention the various connecting devices?
61. What is the uses of bridges?
62. What is meant by cut-through switching?
63. Define spanning tree? (MAY 2016)
64. What is meant by Repeaters?
65. What is meant by hub? (MAY, NOV 2016)
66. What is meant by bridges?
67. What is meant by Switches? (NOV 2016)
68. What is meant by routers?
69. What is meant by gateway?
70. What are Virtual LANs? (NOV 2015)
71. Differentiate between physical and logical address. (MAY 2017)

11 MARKS

1. Explain in detail about design issues in data link layer?
2. Explain in detail about Error Detection and Error Correction codes? (NOV 2015) (MAY 2016, 2017)
3. Explain in detail about elementary data link layer protocols? (6 MARKS)
4. Explain in detail about simplex protocol in data link layer? (5 MARKS)
5. Explain in detail about Stop and Wait protocol? (MAY 2017)
6. Explain in detail about Sliding Window Protocols? (NOV 2016)
7. Explain in detail about Medium Access Control Sub-layer?
8. Explain the ALOHA in Multiple Access Protocol?
9. Explain in detail about CSMA protocols? (NOV 2015)(MAY 2016)
10. Explain the various Collision-Free Protocols?
11. Explain in detail about Limited-Contention Protocols? (5 MARKS)
12. Explain in detail about Wireless LAN 802.11 Protocol architecture?
13. Explain MAC sub layer protocol and frame structure of IEEE 802.11? (NOV 2016)
14. Explain in detail about Wireless LAN 802.16 Protocol architecture? (6 MARKS)

15. Explain about the uses of bridges and learning bridges using Data link layer switching?(6 MARKS)
16. Write a short notes on Spanning Tree Bridges? (5 MARKS)(NOV 2015)
17. Explain in detail about Repeaters, Hubs, Bridges, Switches, Routers, and Gateways? (NOV 2015)
18. Explain in detail about Virtual LANs? (6 MARKS)

UNIT-III

2 MARKS

1. What is meant by Network layer?
2. Mention the various issues in network layers?
3. What are the Services provided to the Transport Layer?
4. What is meant by datagram?
5. What is meant by virtual circuit network?
6. What is meant by packet switched network?
7. Comparison between datagram network and virtual network?
8. What is routing algorithm?
9. What is mean by session routing?
10. What is forwarding?
11. Mention two major classes in routing algorithm?
12. What is Adaptive and Non-adaptive routing algorithm?
13. What is static and dynamic routing?
14. List the variety of routing algorithms?
15. Define sink tree?
16. Describe the shortest path algorithm?
17. Define flooding?
18. What is mean by distance vector routing?
19. State the purpose of maintaining the routing table and where is it maintained? (MAY 2017)
20. What is mean by Link state routing?
21. Mention the five approaches in Link state routing?
22. Define OSPF?
23. What is mean by Hierarchical Routing?
24. What is mean by broadcasting routing?
25. Define multi-destination routing?
26. What is reverse path forwarding?
27. What is spanning tree?
28. Define multicast routing?
29. Define MOSPF?
30. Define DVMRP?

31. Define CBT?
32. Define PIM?
33. What is mean by Anycast routing?
34. Difference between connection-less and connection-oriented service?
35. Define congestion?
36. What is congestion control?
37. Mention the various approaches in congestion control?
38. What is network provisioning?
39. What is Traffic-aware routing?
40. What is admission control?
41. What is load shedding?
42. Why is admission control required in networks? (NOV 2015)
43. Why traffic shaping approach is used in congestion. (MAY 2016)
44. Define choke packets?
45. Define RED?
46. Define internetworks?
47. List the many ways networks can differ?
48. How the networks can be connected?
49. What is multi-protocol router?
50. What is mean by Tunneling?
51. Mention the two-level internetworking routing algorithm?
52. Define autonomous system?
53. Define Path MTU?
54. Mention the two types of fragmentation?
55. Define path MTU discovery?
56. What is mean by IPv4?
57. What is Subnetting?
58. State the difference between classless and classful addressing. (NOV 2016)
59. What is mean by IPv6?
60. Bring out the difference between IPv4 and IPv6 address? (NOV 2015)
61. What is mean by NAT?
62. List the various IPv6 extension header?
63. Define ICMP?
64. List the types of ICMP messages?
65. Define DHCP?
66. What is OSPF?
67. List the five types of OSPF messages?

68. Define BGP?

69. Find the error, if any in the following IP addresses. (NOV 2016)

(a) 111.56.045.78 (b) 75.45.301.14

11 MARKS

1. Explain in detail about the design issues in network layer? (MAY 2016)
2. Explain in detail about Routing Algorithm? (6 MARKS)
3. Discuss the optimality principle of routing? (6 MARKS) (NOV 2015)
4. Explain in detail about Shortest Path Algorithm? (6 MARKS)
5. Write a short notes on Flooding? (5 MARKS)
6. Explain the working of Distance Vector Routing? (NOV 2015, 2016) (MAY 2017)
7. Explain briefly about Link State Routing? (NOV 2016)
8. Write a short notes on Hierarchical Routing? (5 MARKS)
9. Write a short notes on Broadcast Routing?
10. Explain briefly about Multicast Routing?
11. Explain in detail about Congestion Control Algorithms? (6 MARKS)
12. Explain the various approaches in congestion control algorithm? (NOV 2015)
13. Explain briefly about Internetworking and also explain the networks differ? (MAY 2016)
14. Write a short note on Tunneling? (5 MARKS)
15. Write a short notes on Internetwork Routing? (5 MARKS)
16. Explain in detail about Packet Fragmentation? (MAY 2016)
17. Explain briefly about IPv4 in detail? (6 MARKS) (NOV 2016)
18. Explain briefly about IP Addresses?
19. Write a short notes on IPv6? (6 MARKS) (NOV 2016)
20. Explain in detail about Internet Control Protocol?
21. Explain the following internet control protocols. (NOV 2015, 2016)
(a) OSPF (b) BGP

COMPUTER NETWORK IMPORTANT QUESTIONS

Rule 1: Use many colour pens to Write the exams

Rule 2: Read all the diagrams

Rule 3: Read important points and Keywords. Don't read full questions.

Rule 4: Each 11 mark should be 6 pages. So read accordingly.

Rule 5: No stories should be written. Only relevant wordings and Keywords.

Unit 1

1. Reference models Page 14 to Page 20
2. Transmission media Pages 37 to Page 45
3. Wireless Transmission page 46 to Page 53

Unit 2

1. Error detection and correction methods page 9 to Page 17
2. Sliding window Page 26 to Page 32
3. CSMA protocol Page 39 to Page 41
4. Data Link layer Protocol page 17 to Page 25
5. Bridges, repeaters, hubs, switches, routers, Gateway's – Read from internet for 2 marks .

Unit 3

1. Optimality Principle of routing page 38 to Page 41
2. Distance vector routing page 11 to Page 14
3. Congestion control Page 22 to Page 25
4. OSPF page 46
5. Packet Fragmentation Page 38 to Page 41
6. BGP protocol read from internet.

Unit-4

1. Flow control and Buffering page11 to page12
2. Transport protocol UDP page 12 to Page 14
3. Transmission control Protocol TCP page14 to page18

Unit 5

1. E mail architecture page 4 to Page 9
2. WWW architecture page 9 to Page 12
3. RSA algorithm page 26 to Page 28
4. Cryptography page 20 to Page 26